

Total No. of Questions : 8]

SEAT No. :

PC-2634

[Total No. of Pages :2

[6354]-783

B.E. (Artificial Intelligence and Data Science)
QUANTUM ARTIFICIAL INTELLIGENCE
(2019 Pattern) (Semester - VII) (Elective - III) (417523 (A))

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Solve questions Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Assume suitable data if necessary*
- 4) *Figures to the right indicate full marks.*

Q1) a) Explain Quantum Part of the Shor's Algorithm. **[9]**

b) Describe Grover's Algorithm. **[9]**

OR

Q2) a) Explain Factoring Integers, with example. **[9]**

b) Discuss analysis of Deutsch algorithm in details. **[9]**

Q3) a) Explain Nearest Neighbour Search in detail. **[9]**

b) Explain Quantum Algorithms for Linear Algebra. **[9]**

OR

Q4) a) Identify the key differences between classical SVM and Quantum SVM in terms of their advantages? **[9]**

b) Explain quantum boosting with example. **[9]**

Q5) a) Explain Quantum Teleportation in detail. **[9]**

b) Describe Quantum cryptography and secure communication in detail. **[8]**

OR

P.T.O.

- Q6)** a) Explain in short Quantum Dense Coding. [9]
b) Explain Noise and error models in quantum systems. [8]

- Q7)** a) State the challenges and limitations are associated with quantum computing in data sciences. [9]
b) Describe Heuristic Search with example. [8]

OR

- Q8)** a) Describe the Quantum Walk-Random Insect. [9]
b) Write a short note on Quantum Tree Search. [8]

